Tax questions are simple to ask, hard to answer correctly

In plain words: people pay a CA or trust a random blog for routine questions — “old or new regime?”, “GST on an AC?”, “do I need GST registration?”

Why now: the law changed twice in a year, so the right answer depends on the date — and chatbots answer from outdated memory, without sources, and get arithmetic wrong.

Our goal: answer in English or Hinglish, cite the official rule for every sentence, use the law in force on the asked date, compute with a calculator, and say “ask a CA” when unsure.

Two law changes

22 Sep 2025

GST 2.0

4 slabs → 5% / 18% (+40% sin goods). AC: 28% → 18%

1 Apr 2026

Income-tax Act, 2025

Replaces the 1961 Act. “Tax Year”; 80C → s.123

Income tax: two regimes (Tax Year 2026-27)

	Old regime	New regime (default)
Idea	High rates, many deductions	Low rates, few deductions
Slabs	0 / 5 / 20 / 30% (from ₹2.5 L)	0 / 5 / 10 / 15 / 20 / 25 / 30% (from ₹4 L)
Deductions	80C, 80D, HRA, home loan…	Standard deduction + employer NPS
Std. deduction	₹50,000	₹75,000
Zero tax up to	₹5 L	₹12 L (₹12.75 L salary)

Effect on Indians: most taxpayers moved to the new regime (~88% for AY 2025-26, per CBDT Chairman). Top real confusion on Reddit: “no tax up to ₹12 L — but 5% from ₹4 L?” (both true: slabs, then the rebate cancels it).

RAG: the LLM writes the answer, but never decides the facts

LLM alone ("closed book")

Answers from memory frozen at training time

✗ outdated (pre-2025 slabs, 28% GST on AC)

✗ no sources ✗ arithmetic errors

Measured (B0): only **10%** of key facts right

RAG ("open book")

Search a trusted, dated library first; paste the passages into the prompt

✓ current law for the asked date

✓ every sentence cited ✓ calculator for numbers

Measured (C6): **69%** key facts, 100% valid citations

Scope delivered (M1 → M3)

M1 · Problem

personas, business case, success metrics fixed before building

M2 · Data

107-passage dated library, 176-question gold set, 80 real questions, leakage-safe split

M3 · Models

baseline B0 + 7 systems B1→C6, all measured on the same dev set, logged in MLflow

What the user gets

Cited answers
source chip on every sentence

Period-aware
law in force on the asked date

Hinglish
"kitna hai?" works

Old section numbers
80C → s.123

Regime calculator
salary or freelancer, ₹ or US$

Safe refusal
"ask a CA" when unsure

Offline: build a dated library. Online: 12 fixed steps for every question

OFFLINE — once

Official sources
CBDT, CBIC, GST Council

→ **1 rule = 1 passage**
107 passages

→ **Tag every passage**
law, section, dates, URL

→ **Two indexes**
BM25 words + e5 vectors

ONLINE — every question (config C6)

1. **PII scrub** — mask PAN/phone
2. **Hinglish → English** — dictionary
3. **Intent** — TF-IDF + LogReg
4. **Period** — dates → law
5. **Entities** — 80C, ₹, $/month
6. **BM25 search** — exact words
7. **Dense search** — e5 meaning
8. **RRF merge** — 1/(60+rank)
9. **Date filter** — swap in current version
10. **Safety gate** — reranker score ≥ θ
11. **Calculator / LLM** — Qwen2.5-3B writes
12. **Citation check** — drop unsupported

Design rule: everything that must be correct — which law, which numbers, which source, answer-or-refuse — is decided by testable components before the LLM runs. The LLM only turns 3 passages into plain sentences, and the citation check deletes any sentence it can't back.

“Naye regime mein standard deduction kitna hai?”

2 Hinglish	naye→new · mein→in · kitna→how much · hai→is ⇒ “New regime in standard deduction how much is?”	2 ms
3 Intent	TF-IDF word 1-2-grams + char 2-5-grams → LogReg: question 0.936 · calculation 0.048 · out-of-scope 0.016	5 ms
4 Period	no date in question → current law: Tax Year 2026-27 (today 29 Oct 2026)	4 ms
6 BM25	tokens [new, regime, standard, deduction, much] → STDDED-1961 7.29 · STDDED-TY2627 6.75	1 ms
7 Dense e5	SentencePiece ▁que ry : ▁New ▁regime … ▁de duction … → 384-d vector · cos: STDDED-1961 0.889 · TY2627 0.883	290 ms
8 RRF	Σ 1/(60+rank) → STDDED-1961 first — both searchers prefer the OUTDATED 1961-Act passage	0 ms
9 Date filter	13 of top-20 out of period removed; each swapped for its current version: STDDED-1961 → STDDED-TY2627	1 ms
10 Reranker gate	XLM-R cross-encoder [CLS] q [SEP] passage → best +3.30 ≥ θ = −5.83 → answer	4.8 s
11 Qwen2.5-3B	~1,200-token prompt (rules + 3 passages + question) → “…standard deduction of ₹75,000 [IT-NEW-STDDED-TY2627]”	18.3 s
12 Citation check	1 tag, 1 valid, 0 sentences dropped · disclaimer added	0 ms

Total ≈ 23 s on a 4-core CPU — the LLM (18 s) and reranker (5 s) dominate. Each model tokenises differently: BM25 sees whole words; e5 and the reranker see XLM-R sub-words (250k vocab: “deduction” = ▁de + duction); Qwen uses byte-level BPE (~151k vocab).

Eight components — only one of them is a generative model

Role	Model / tool	Size	Why
Keyword search	BM25 (rank_bm25)	no training	Exact terms: 80C, ITR-4, “August 2025”
Embeddings	intfloat/multilingual-e5-small	118M · 384-d	Meaning + Hinglish; ‘query:’/‘passage:’ prefixes, cosine
Fusion	Reciprocal Rank Fusion	k = 60	Merge rankings without calibrating scores
Reranker → gate	BAAI/bge-reranker-base	278M · XLM-R	Answer-or-refuse confidence (θ = −5.83, tuned on dev)
Intent	TF-IDF + Logistic Regression	342 rows · CV 0.90	question / calculation / out-of-scope in ~5 ms
Rules	Hinglish dict · dates · entities	~370 words	Exact, testable; every replacement shown
Calculator	Deterministic Python	16/16 exact	Both regimes, 3 years, rebate, cess, s.58, FX
Writer LLM	Qwen2.5-3B-Instruct (Ollama)	3.1B · Q4_K_M	Temp 0, top-3 passages, must cite every sentence

Serving

FastAPI + Uvicorn
/ask /compare /calculate /experiments

Vanilla HTML/CSS/JS
live pipeline trace, charts

MLflow
every config logged

pytest
101 tests

VPS, CPU only
4 vCPU · 8 GB · no GPU

Tailscale Funnel
public HTTPS link

One change per step, same LLM, same 53 dev questions

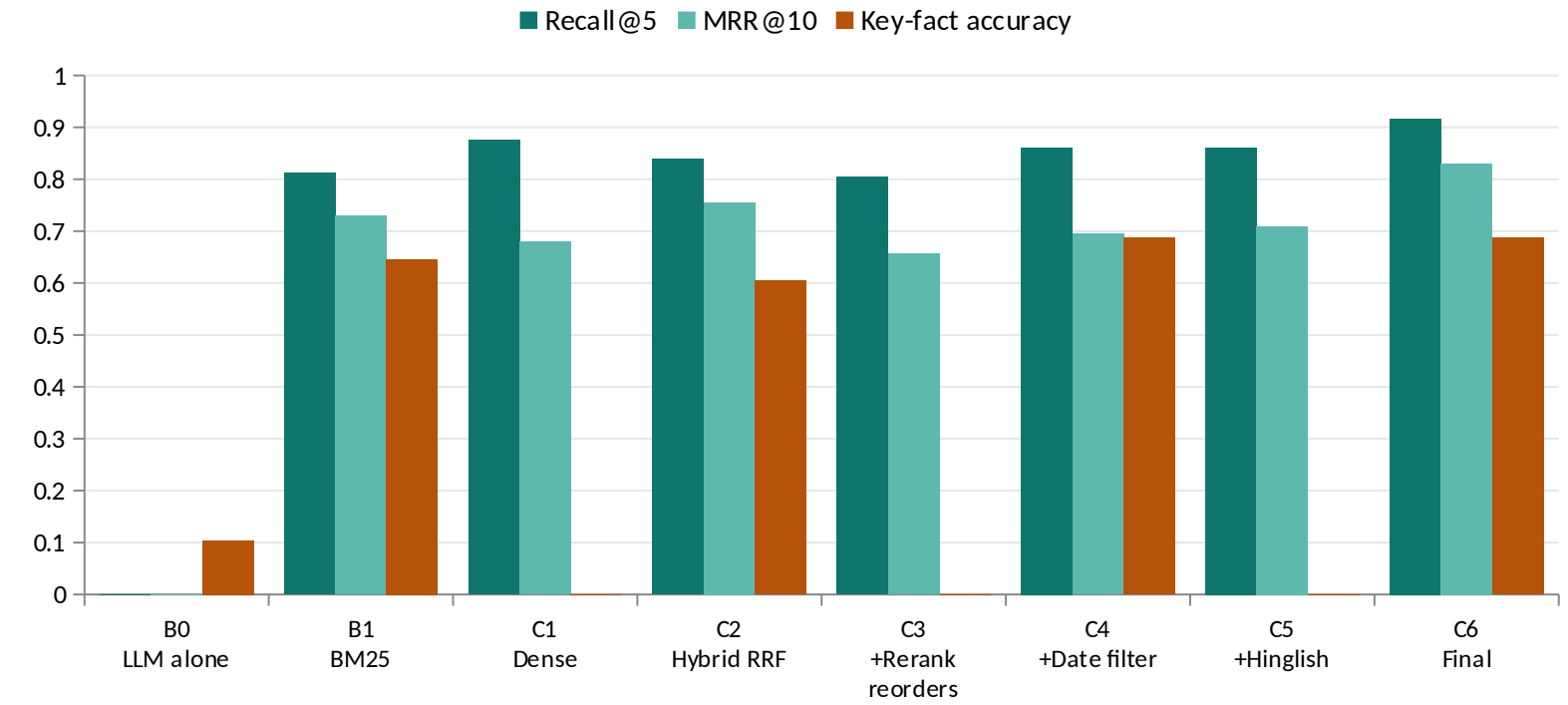
M2 · Data

107 passages
17 official sources · 72 verified on official pages

176 gold questions
income tax 60 · GST 42 · time traps 26 · calc 18 · out-of-scope 18 · Hinglish 12

Split 53 dev / 123 test
stratified, seed 42 — test untouched until M4 (no leakage)

80 real questions
Reddit, Quora, CA forum, official FAQs — never used for tuning



Key-fact accuracy needs the LLM (~25 s/answer on CPU), so it was run for B0, B1, C2, C4, C6; 0 = not run.

Key facts
10% → 69%
LLM alone → full system

Period accuracy
88% → 100%
date filter (C2 → C4)

MRR
0.755 → 0.656
reranker reordering hurt (C3)

C6 final
R@5 0.917 · MRR 0.830
best on both

What worked, what didn’t, and where the LLM belongs

✓ **Retrieval is essential**
key facts 10% → 69%; the LLM’s memory predates the 2025–26 law changes

~ **Dense alone: better recall, worse dates**
C1 Recall@5 0.875 but period accuracy 75%: “Aug 2025” ≈ “now” in vector space

✓ **Hybrid RRF ranks best (single stage)**
C2 MRR 0.755: BM25 exact tokens + dense paraphrase

✗ **Reranker reordering hurt**
MRR 0.755 → 0.656; EN/ZH-only model promotes related or wrong-period passages

✓ **Fix: reranker as confidence gate only (C6)**
best Recall@5 0.917 and MRR 0.830

✓ **Date filter + period substitution**
period accuracy 100%; swaps an outdated passage for the same rule’s current version

Limits / next (M4): curated summaries not full Acts · team-written gold set · real out-of-scope refusal 57% · run the untouched test set · GPU for speed

Targets fixed in M1 (before building) — C6

Recall@5 ≥ 0.85	0.917 ✓
Faithfulness ≥ 0.90	1.000 ✓
Out-of-scope refused ≥ 90%	100% ✓
Calculator exact = 100%	100% ✓
Median latency ≤ 5 s	16.8 s (CPU) ✗

Why an LLM — and only at the end?

It is the only part good at turning 3 legal passages into 2–5 plain sentences for this user. Everything it is bad at (dates, numbers, sources, knowing when to stop) is handled before it, by testable code. Small local model (Qwen 3B): free, private, reproducible — the cost is ~17 s on CPU.